

QUANTUM GRAVITY — FEATURE

One Field to Rule Them All: The Physicist Who Says Everything Is Made of the Same Stuff

Dustin Lee's Unified Substrate Theory proposes that spacetime, matter, and quantum mechanics are all different faces of a single underlying field — and he has thirteen experiments that might prove it.

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Somewhere above the Louisiana marshland, a laser beam is measuring the universe. It travels four kilometers down a hollow steel tube, bounces off a mirror polished to atomic smoothness, returns, and interferes with a second beam at a precision that would have seemed delusional to physicists a generation ago. The LIGO detector — the Laser Interferometer Gravitational-Wave Observatory — is sensitive enough to detect a distortion of spacetime one ten-thousandth the diameter of a proton. It has been doing this, on and off, for a decade. And it has been finding things.

What it has not found, yet, is the thing that would upend physics entirely: a gravitational wave that arrives a fraction of a second later than it should, or that carries an extra polarization mode that general relativity forbids, or that echoes faintly after the main signal, like a rumble following thunder. Those would be signs that spacetime itself is not what Einstein thought it was. That beneath the smooth, curved fabric of general relativity, something else is going on.

Dustin Lee believes he knows what that something is. He calls it the substrate.

Lee, a theoretical physics researcher based in Walstonburg, North Carolina, has spent years developing what he calls the Unified Substrate Theory — UST for short — a framework in which spacetime, matter, energy, and quantum behavior are not separate phenomena requiring separate explanations. They are, in his account, all expressions of a single primordial field that underlies everything. Not a field in the ordinary physics sense of a quantity that varies through space. Something more fundamental. Something that generates space, generates time, generates particles, generates the rules by which particles behave. The substrate comes first. Everything else is downstream.

It is an audacious claim. Physics is not short of audacious claims. What makes UST worth serious attention is this: Lee has written down thirteen specific, falsifiable predictions that distinguish his framework from general relativity and from quantum mechanics as currently understood — and several of those predictions are being tested right now, by instruments already operating.

We do not yet know what those instruments will find. That is the nature of science. But the experiments are running.

The Problem at the Heart of Physics

To understand why Lee's proposal matters, it helps to appreciate the strange and uncomfortable position that modern physics occupies. For most of the twentieth century, physicists built two extraordinary theoretical frameworks — frameworks so successful, so precisely verified, that they represent the greatest intellectual achievements in the history of science. And these two frameworks, at a fundamental level, are incompatible with each other.

The first is general relativity, Einstein's 1915 theory of gravitation. In Einstein's picture, gravity is not a force. It is geometry. Mass and energy warp the fabric of spacetime, and other objects follow the curves of that warped geometry. A planet orbits a star not because a force is pulling it, but because it is following the straightest possible path through curved spacetime. General relativity is a theory of smooth curves, of continuous fields, of a spacetime that can be bent and stretched but that is, at every point, precisely defined.

The second framework is quantum mechanics — and its mature descendant, quantum field theory. In this picture, the world at small scales is fundamentally different. Quantities that seem continuous are actually discrete. A particle does not have a definite position and momentum simultaneously; it exists in a superposition of possibilities until a measurement forces it to choose. Forces are mediated not by smooth fields but by the exchange of quantum particles — photons for electromagnetism, W and Z bosons for the weak force, gluons for the strong force.

Quantum field theory is, in terms of predictive precision, the most successful scientific theory ever constructed. Its predictions have been verified to better than one part in a trillion. General relativity, tested across systems ranging from GPS satellites to merging black holes a billion light-years away, has never been wrong by even a measurable amount.

And yet.

The two theories cannot both be completely right. Not at the deepest level. Quantum field theory requires a fixed spacetime background — a stage on which particles perform. General relativity says the stage itself is dynamic, curved, affected by what is on it. There is no agreed mathematical procedure for marrying the quantum description of the world to a dynamic spacetime. When physicists try to calculate quantum gravity effects by the usual methods — writing down a graviton, the hypothetical quantum of gravity, and computing

interactions — the equations blow up. Infinities appear that cannot be tamed by the mathematical procedures that work everywhere else in quantum field theory. The theory is not renormalizable. It breaks down.

This is not a minor technical inconvenience. It is a sign that something fundamental is missing. The breakdown is most acute in two physical situations: the interior of black holes, where matter is compressed to infinite density, and the first instant of the Big Bang, where the entire observable universe was compressed to smaller than a proton. In both cases, the physics of both general relativity and quantum mechanics becomes relevant simultaneously, and both theories, individually, produce nonsense.

The correct theory — the theory that works in both regimes, that reduces to general relativity where gravity dominates and to quantum mechanics where it doesn't — does not yet exist. Finding it is arguably the central open problem in all of fundamental physics.

“The two most successful theories in the history of science cannot both be completely right. At some deep level, one of them is telling us a story that eventually runs out.”

— On the GR-QM incompatibility

Where the Others Fell Short

This is not the first time someone has tried to solve the problem. For half a century, some of the most brilliant minds in theoretical physics have devoted their careers to quantum gravity. Their track record is instructive.

String theory emerged in the 1970s and reached its zenith of cultural dominance in the 1980s and 1990s. The core insight is elegant: replace point particles with tiny vibrating one-dimensional strings, and the different modes of vibration produce different particles — including, crucially, a spin-2 particle that behaves exactly like the graviton. String theory naturally produces quantum gravity. It is, in that narrow sense, a genuine achievement: the first framework to suggest a mathematically consistent quantum description of gravitation.

But string theory's promise came tethered to a profound complication. The theory requires extra spatial dimensions — typically ten or eleven — that are not observed. To make contact with the four-dimensional world we inhabit, those extra dimensions must be compactified — curled up so small they are invisible. The trouble is that there are an astronomically large number of ways to do this compactification. Each produces a different physics. The number of distinct string theory vacua has been estimated at 10 to the power of 500. This is the "landscape problem." String theory does not predict our universe; it permits it, along with an almost uncountable number of others. Extracting a unique, testable prediction from this landscape has proven essentially impossible, and string theory has generated essentially no confirmed experimental predictions in fifty years of effort.

Loop quantum gravity, developed principally by Lee Smolin and Carlo Rovelli beginning in the 1980s, took a different approach. Instead of adding extra dimensions or new objects, it quantized spacetime itself. In loop quantum gravity, space is not continuous. It is made of discrete quantum chunks — "spin networks" — with a minimum size near the Planck length, approximately 10 to

the power of negative 35 meters. This produces a mathematically rigorous quantization of the gravitational field that does not require extra dimensions or exotic new physics.

Loop quantum gravity's achievement is real. It offers a picture of quantum spacetime that is internally consistent, and it has generated some genuine predictions — notably, modifications to the dispersion relation of photons that might be detectable using gamma-ray bursts. But it has struggled to fully incorporate matter. The Standard Model particles — quarks, leptons, bosons — do not emerge naturally from the spin network framework. They must be added by hand, imported from elsewhere. The theory is a quantum theory of gravity, but it is not yet a quantum theory of gravity-plus-everything-else. The matter sector remains incomplete.

Emergent gravity proposals — the idea that gravity is not fundamental but emerges from some deeper structure, perhaps as an entropic force or as a collective effect of quantum information — have generated intense interest, particularly following work by Erik Verlinde. But these frameworks have their own gap: they typically assume a pre-existing geometric arena within which emergence occurs. They do not explain where that arena comes from. The pre-geometric question — what is the stuff before spacetime — is left open.

Lee's theory is, in part, a direct response to these gaps. The landscape problem of string theory arises because the theory offers too much freedom. The matter incompleteness of loop quantum gravity arises because gravity and matter are still treated as distinct. The pre-geometric gap in emergent gravity arises because something is assumed to exist before the geometry emerges. UST attempts to close all three gaps with a single move: posit one thing, from which everything else follows.

“String theory gave us too many universes. Loop quantum gravity gave us space but not matter. Emergent gravity forgot to explain what was doing the emerging. Lee’s proposal is a response to all three.”

— The state of quantum gravity, circa 2026

Enter the Substrate

Lee’s starting point is deceptively simple. What if there is one thing?

Not one force, not one particle, not one symmetry — but one field. A single, primordial, universal field that exists at a level deeper than spacetime itself. Lee calls it the substrate. It is not made of anything else. It does not live in spacetime. Rather, spacetime lives in it. Or, more precisely, spacetime emerges from it.

The analogy he reaches for is the ocean. Imagine you are a physicist who has only ever studied waves. You have developed extraordinarily precise theories of wave behavior: how waves propagate, how they interfere, how they break on shores, how their frequencies and amplitudes are related. Your wave theory is spectacularly successful. And then someone tells you that the waves are not fundamental. The water is. The waves are just the water doing something.

That is roughly the logical structure of UST. General relativity and quantum mechanics are the wave theories. The substrate is the water. Both theories work beautifully in their respective domains because they are correct descriptions of the behavior of the substrate in different regimes. But neither theory is describing the substrate itself. Neither theory is the deepest story.

So what is the substrate doing? According to Lee's framework, the substrate is a continuous, dynamic field with certain intrinsic properties. Where it is smooth and gently curved, it produces what we experience as empty spacetime. The curvature of the substrate is what we measure as gravity. A gravitational field is not a separate thing added to spacetime — it is the substrate itself curving. Einstein was right that gravity is geometry, but the geometry belongs to the substrate, not to spacetime as a fixed background.

Particles — electrons, quarks, photons — are something else entirely. They are not little balls. They are not even point-like excitations of separate quantum fields, as the Standard Model describes them. In UST, particles are topological defects in the substrate. They are knots.

Think of a rope. You can run your hand along the rope and it feels smooth and uniform. But if you tie a knot in it, something changes. The knot has a definite location. It has a definite character — the topology of the knot, the way it is tied. You can move the knot along the rope, but you cannot easily untie it without threading the end of the rope back through. The knot is, in a sense, a stable, localized feature of the rope that behaves differently from the rope's background.

In UST, an electron is a knot. A specific topological configuration of the substrate field. The electron's charge, mass, and spin are not properties it carries with it like a name tag. They are properties of the knot topology itself. Different particles are different kinds of knots — different topological classes, different configurations, each with their characteristic properties determined entirely by their topology.

Forces, in this picture, are gradients. When two electrons approach each other, the topology of their respective knots distorts the local substrate field, creating a gradient — a variation in the field's configuration across space. That gradient is the electromagnetic force. The particles do not exchange photons in the first

instance; the photon, too, is a particular kind of excitation propagating through the substrate, and the exchange picture is an approximation that works extraordinarily well at ordinary energies but breaks down at the deepest level.

Quantum behavior — the uncertainty principle, superposition, wave-particle duality — is not, in this framework, a mysterious feature of the world that must be axiomatically accepted. It is the natural behavior of the substrate at small scales. When you probe the substrate at length scales near its minimum granularity — near the Planck scale — you encounter the substrate's intrinsic discreteness and fluctuation. That intrinsic behavior is what we call quantum mechanics. It is not separate from gravity. Both are the substrate. One is the substrate at large scales. The other is the substrate at small ones.

The Central Claim of UST

Spacetime is not fundamental — it emerges from the substrate's large-scale behavior.

Particles are topological defects (knots) in the substrate field, not separate quantum objects.

Forces are gradients in the substrate configuration, not exchanged particles at the fundamental level.

Quantum behavior is the substrate's natural small-scale dynamics — not a separate axiom system.

Gravity is the substrate curving — not a force and not mediated by a particle.

What the Theory Actually Says

Lee is not content with philosophy. A framework that cannot make quantitative predictions is not a scientific theory — it is metaphysics. UST makes quantitative predictions. To understand them, it helps to look briefly at the mathematical heart of the proposal.

The dynamics of any classical field theory are encoded in a mathematical object called the Lagrangian — a function that, when minimized, yields the equations of motion. General relativity has its Lagrangian (the Einstein-Hilbert action). Quantum electrodynamics has its Lagrangian. UST has a modified one. In symbolic form:

$$\mathcal{L}$$
$$=$$
$$\mathcal{L}$$
$$\text{GR}$$
$$+$$
$$\alpha$$
$$\cdot$$
$$R$$
$$\mu\nu\rho\sigma$$
$$R$$
$$\mu\nu\rho\sigma$$
$$+$$
$$\mathcal{L}$$

matter

(substrate)

The UST Lagrangian: Einstein-Hilbert gravity, a higher-derivative correction term (the Kretschner scalar, weighted by constant α), and matter arising from substrate topology.

Do not worry about the Greek letters. The key feature is the middle term: a higher-derivative correction to Einstein's gravity, controlled by a small constant that Lee calls α . At ordinary scales — the scales of planets, stars, and even neutron stars — this correction is negligible. General relativity is recovered. But at extreme energies, near the Planck scale, near black hole singularities, near the Big Bang — wherever the curvature of spacetime becomes enormous — this correction becomes important. It is in those extreme regimes that UST departs from general relativity in measurable ways.

Those departures are what generate Lee's thirteen predictions. Here are the most experimentally acute.

Energy-dependent photon speed. In UST, light does not travel at exactly the same speed at all energies. High-energy photons — gamma rays — travel infinitesimally slower than low-energy photons. The difference is tiny: completely undetectable over the distance from your kitchen light to your eyes, or even across the solar system. But gamma rays that travel across billions of light-years accumulate the discrepancy. If a gamma-ray burst fires high-energy and low-energy photons simultaneously, and you detect them at Earth, the high-energy photons should arrive slightly later. The Fermi Large Area Telescope has been measuring gamma-ray bursts for years with exactly this test in mind. So far, the limits are tight, but not yet tight enough to decisively rule out theories at the level of correction UST predicts.

Gravitational wave echoes. In standard general relativity, when two black holes merge, they produce a characteristic gravitational wave signal: a chirp, rising in frequency and amplitude, followed by a ringdown as the merged remnant settles. In UST, the horizon of a black hole is not a hard, perfectly sharp boundary at the substrate level. The substrate does not abruptly end at the event horizon. This means that gravitational waves can partially reflect off the near-horizon region and return, producing faint echoes after the main signal. Several teams have looked for echoes in LIGO data. Results have been claimed and debated. None have yet been confirmed to the standard required for a detection claim, but the search continues with each observing run.

Anomalous gravitational wave polarization. General relativity predicts that gravitational waves come in exactly two polarization modes — the "plus" and "cross" polarizations. Some extended theories of gravity predict additional modes. UST predicts them too, arising from the substrate's additional degrees of freedom. Detecting these extra modes requires three or more gravitational wave detectors operating simultaneously. The expansion of the detector network — LIGO, Virgo, KAGRA, and the forthcoming Einstein Telescope — is making this test progressively more realistic.

Generalized Uncertainty Principle corrections. Heisenberg's uncertainty principle says you cannot simultaneously know the position and momentum of a particle with perfect precision: the more precisely you know one, the less precisely you can know the other. In UST, because the substrate has a minimum granularity — you cannot have structures smaller than the Planck length — the uncertainty principle acquires a correction term. The modified uncertainty relation looks like:

$$\Delta$$

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \cdot [1 + \beta (\Delta p)^2]$$

The Generalized Uncertainty Principle (GUP): a standard Heisenberg relation, modified by an extra term proportional to the square of the momentum uncertainty. The constant β encodes the substrate's minimum length scale.

At ordinary lab energies, the extra term is negligible — the correction is of order 10 to the power of negative 35, far too small to see in any current experiment. But precision tabletop experiments — notably the Bose-Marletto-Vedral (BMV) experiment and its successors, which aim to create quantum superpositions of mesoscopic masses — are beginning to probe regimes where such corrections might accumulate to measurable levels. This is one of the most carefully watched frontiers in experimental physics.

The full list of thirteen predictions spans multiple experimental platforms, from the cosmic microwave background (probed by CMB-S4, a next-generation array of ground-based telescopes) to ultra-high-energy cosmic ray spectra measured at the Pierre Auger Observatory in Argentina. Each prediction is not merely qualitative — it comes with a specific parametric dependence that distinguishes UST from other quantum gravity proposals.

UST Prediction	Testing Instrument(s)	Status (2026)
Energy-dependent photon speed (Lorentz violation at Planck scale)	Fermi-LAT, CTA	Ongoing; limits approaching UST parameter range
Gravitational wave echoes post-merger	LIGO/Virgo/KAGRA	Active search; no confirmed detection yet
Extra GW polarization modes (beyond +/×)	LIGO, Virgo, KAGRA, Einstein Telescope	Network sensitivity increasing with each run
Generalized Uncertainty Principle (GUP) corrections	BMV experiment, tabletop quantum gravity labs	Next-generation precision measurements in progress
Modified black hole shadow / photon ring structure	Event Horizon Telescope	High-resolution imaging campaigns ongoing
CMB primordial power spectrum modifications	CMB-S4, Simons Observatory	Data collection phase; analysis pipeline in development
Ultra-high-energy cosmic ray spectrum modifications	Pierre Auger Observatory	Accumulating statistics; multi-messenger analysis active
Non-thermal correlations in Hawking radiation	Analog gravity experiments (lab-scale)	Preliminary lab analogs being developed

UST Prediction	Testing Instrument(s)	Status (2026)
Substrate-induced decoherence in quantum systems	Precision quantum optics, atom interferometry	Long-baseline atom interferometers under construction
Minimum length imprint on Hubble tension	DESI, Euclid, JWST	H0 tension measurements ongoing; UST angle being studied
Modified neutron star equation of state	NICER, LIGO neutron star mergers	Data accumulating from X-ray timing observations
Topological charge quantization signatures	High-energy collider experiments	Theoretical analysis of LHC data in progress
Graviton mass bound from substrate dynamics	LIGO GW propagation tests	Tight bounds already set; UST predicts massless graviton equivalent

The Questions It Answers

If UST is correct — a conditional that must be emphasized — it resolves, along the way, a remarkable number of problems that have troubled physicists for decades. Some of these resolutions are elegant enough to be worth examining even before experimental confirmation, because they illustrate what kind of theory UST is trying to be.

The Black Hole Information Paradox. In 1974, Stephen Hawking showed that black holes should radiate — that quantum effects near the event horizon cause a black hole to slowly emit thermal radiation and eventually evaporate. This created an immediate problem: if the radiation is perfectly thermal, it carries no information about what fell in. A star, a library, an alien civilization — if all of them fell into the same black hole and the black hole evaporated, the radiation would look identical. The information would be destroyed. But quantum mechanics says information cannot be destroyed. This contradiction — known as the information paradox — has resisted resolution for fifty years.

In UST, the Hawking radiation is not perfectly thermal. The substrate topology that encodes the matter that fell in leaves faint, non-thermal correlations in the outgoing radiation — correlations too subtle to be detected individually, but which, taken together, encode the complete information about the infalling matter. Unitarity — the quantum mechanical principle that information is conserved — is preserved. The paradox dissolves not because of any new exotic mechanism, but because the horizon is, at the substrate level, not the perfectly sharp, information-erasing barrier that it is in classical general relativity.

The Firewall Paradox. In 2012, physicists Ahmed Almheiri, Donald Marolf, Joseph Polchinski, and James Sully published what became known as the AMPS paper, arguing that if information is preserved, the event horizon must be a region of extremely high energy — a "firewall" — that would incinerate an infalling observer rather than the gentle, unremarkable crossing that general relativity predicts. The paradox pits the equivalence principle (an infalling observer should notice nothing special at the horizon) against unitarity (information must be preserved).

UST's answer is that the firewall paradox is a consequence of treating the horizon as a fundamental geometric object. At the substrate level, the horizon is emergent — a large-scale, coarse-grained approximation to what is actually a smooth region of substrate field. There is no hard geometric boundary at which information is necessarily trapped or at which a firewall must form. The paradox, in this account, arises from taking the classical geometric description too seriously in a regime where it breaks down. At the substrate level, it simply does not arise.

Why Every Electron Is Identical. Physicists call this the "identical particle problem," though it rarely makes it into popular accounts of physics' great mysteries. Here it is: every electron in the universe has exactly the same mass, exactly the same charge, exactly the same spin. Not approximately the same.

Exactly. Why should this be? If electrons were manufactured objects, like cars or coins, you would expect minor variations. They are not manufactured objects. They are something that appears, spontaneously, in a universe that was not designed. Why are they identical?

In UST, the answer is topological invariance. An electron is a particular class of knot in the substrate field. Two knots of the same topological class have identical properties — not because someone made them the same way, but because topology is exact. There is no such thing as an "almost" trefoil knot versus a "perfect" trefoil knot; topologically, it either is or it isn't. The identical particle problem disappears because particles are not objects that could vary — they are topological invariants.

ER = EPR. In 2013, Juan Maldacena and Leonard Susskind proposed a striking conjecture, abbreviated ER=EPR: that two entangled particles (connected by Einstein-Podolsky-Rosen quantum correlations, or EPR) are connected by an Einstein-Rosen bridge (a wormhole, or ER bridge). The conjecture suggests that quantum entanglement and spacetime geometry are not separate phenomena but two descriptions of the same thing. It is a beautiful idea, but in the framework where it was proposed, it remains a conjecture — motivated by analogy and duality arguments, not derived from first principles.

In UST, ER=EPR is not a conjecture. It is a direct structural consequence of the theory. Entangled particles share a real, physical connection through the substrate — a topological link in the underlying field. What quantum mechanics describes as non-local correlations between measurement outcomes is, at the substrate level, a genuine connection. The wormhole is not a metaphor. It is what entanglement actually is, described at the substrate level rather than at the emergent spacetime level.

Why There Is No Graviton. The Standard Model describes all forces as mediated by exchange particles. Electromagnetism has the photon. The weak

force has W and Z bosons. The strong force has gluons. The natural expectation is that gravity has the graviton — a spin-2 quantum particle mediating gravitational attraction. Physicists have searched for a consistent quantum field theory of the graviton for decades. It cannot be done, at least not by the usual methods: the theory is non-renormalizable.

In UST, the reason there is no graviton — no consistent particle mediating gravity — is that gravity is not a force mediated by particle exchange. Gravity is the substrate curving. At the substrate level, it is geometry. The curvature propagates as waves — gravitational waves, which are real and have been detected — but these are not mediated by particle exchange. They are the substrate itself oscillating. Trying to quantize gravity by finding a graviton is, in UST's account, like trying to find the particle that mediates the fact that space has three dimensions. You will not find it, because the concept is category error. Gravity is not a force. It is the structure of the substrate.

“In UST, ER equals EPR is not a conjecture. It is a structural fact. Entangled particles share a real substrate connection. The wormhole is what entanglement actually is.”

— On quantum entanglement and spacetime topology

What's Still Missing

Good journalism requires honesty, and honesty about UST requires acknowledging what the framework does not yet have.

Lee is forthright about this. These are not fatal objections — they are the normal condition of a developing theoretical framework. Every major theory in physics has, at some stage in its development, had significant gaps. General relativity was proposed before the experimental tests that confirmed it. Quantum electrodynamics was mathematically problematic for years before renormalization resolved its infinities. But the gaps should be named, and examined.

The substrate action is not fully specified. The Lagrangian displayed earlier — the mathematical heart of the theory — contains the higher-derivative correction term, but the precise form of the full substrate action, and particularly the form of the matter sector arising from substrate topology, has not been written down completely. This matters because the exact form of the action determines the exact numerical predictions. Without it, some predictions can only be made at the qualitative level — "there will be echoes" rather than "the echoes will arrive 0.3 seconds after the main signal with amplitude X."

The Standard Model spectrum has not been derived. UST asserts that particles are topological defects in the substrate, and that different topological classes correspond to different particles. This is a compelling picture. It is not yet a complete derivation. The Standard Model contains a specific zoo of particles — six quarks, six leptons, four gauge bosons, and the Higgs — with specific masses, charges, and quantum numbers. A complete UST must derive this spectrum from first principles of substrate topology. That derivation has not been produced. The topological picture provides a conceptual framework for why such a derivation should exist, but the detailed mathematics remains to be worked out.

The cosmological constant is not numerically predicted. The cosmological constant — the energy density of empty space, responsible for the universe's accelerating expansion — is one of the most puzzling numbers in physics.

Quantum field theory predicts it should be 10 to the power of 120 times larger than what is observed. This discrepancy is the largest in all of science. UST addresses the cosmological constant conceptually — it arises from the background energy density of the substrate — but without the complete substrate action, the precise numerical value cannot be calculated. The prediction remains qualitative.

These gaps are genuine, and they represent real work that remains to be done. Lee does not deny them. They also do not undermine the theory's testability at the level of its current predictions — the thirteen observational signatures are sufficiently specified to be falsifiable. But a complete theory must ultimately close these gaps.

Acknowledged Open Questions in UST

- 1. Substrate Action:** The complete mathematical form of the substrate Lagrangian, including the full matter sector, is not yet written down.
- 2. Particle Spectrum:** Full derivation of the Standard Model particle zoo from substrate topology is incomplete.
- 3. Cosmological Constant:** Precise numerical prediction requires the complete action; current treatment is qualitative.

These are the work of future development — not arguments against the framework's coherence or testability at current levels.

The Bigger Pattern

Whatever one thinks of the specific details of UST — and the details are what the experiments will ultimately adjudicate — it is worth stepping back and noticing the pattern in which it participates.

In 1865, James Clerk Maxwell wrote down his equations of electromagnetism. Before Maxwell, electricity and magnetism were two separate phenomena — studied separately, governed by separate laws, produced by separate sources. After Maxwell, they were one thing: two faces of a single electromagnetic field. What seemed to be two phenomena was one. The universe had fewer ingredients than we thought.

In 1905 and 1915, Albert Einstein unified space and time. Before Einstein, space was the arena where events occurred, and time was the universal clock that ticked at the same rate everywhere. After Einstein, space and time were aspects of a single entity — spacetime — that could be bent, stretched, and traveled through at varying rates depending on velocity and gravity. What seemed to be two separate dimensions of reality was one. Again, fewer ingredients.

In the 1960s and 1970s, the electroweak unification of Sheldon Glashow, Abdus Salam, and Steven Weinberg merged electromagnetism and the weak nuclear force into a single electroweak force, predicted the W and Z bosons, and was vindicated in 1983 when those particles were discovered at CERN. Again: what seemed to be two was one.

The pattern is unmistakable. The history of physics is a history of realizing that things we thought were different are the same thing described differently. Lee's proposal is the next step in that sequence. If Maxwell unified electricity and magnetism into one field, and Einstein unified space and time into one manifold, UST proposes to unify the manifold and the fields that live on it into one substrate. Spacetime and matter — not two categories, but one.

This is not proof that UST is right. The history of physics also contains many bold unifications that did not survive contact with experiment. What the historical pattern does tell us is that the direction — toward unification, toward fewer fundamental ingredients — is the proven direction of scientific progress. The question is not whether such a unification exists. The question is whether Lee has found it.

*“Maxwell: electricity and magnetism — one field.
Einstein: space and time — one manifold. Lee: the
manifold and the matter that fills it — one substrate. The
direction of this sequence is the proven direction of
physics.”*

— The unification lineage

The Waiting Game

The instruments do not care about elegance. They do not care about history, or pattern, or the satisfying tidiness of a framework in which everything is one thing. They measure what they measure, and what they measure is what is true.

Right now, as this article goes to press, the LIGO-Virgo-KAGRA network is collecting data in its O4 observing run — the most sensitive gravitational wave search ever conducted. Analysts are combing through the data for echoes. The Fermi Large Area Telescope is measuring gamma-ray bursts from across the observable universe and timing the arrival of high-energy photons against low-energy ones. The Event Horizon Telescope collaboration is refining its image of

the shadow of M87* and Sagittarius A*, looking for deviations in photon ring structure that no standard theory predicts. The CMB-S4 project is moving toward first light. The Pierre Auger Observatory accumulates ultra-high-energy cosmic ray events, one by one, building a statistical picture of the highest-energy particles in the universe.

All of these experiments — designed for different primary purposes by different collaborations on different continents — are, collectively, closing in on the energy and sensitivity scales where theories like UST make their most distinctive predictions. Not because anyone designed them to test UST specifically. Because the frontier of physics has arrived, in 2026, at a place where the deep structure of spacetime is beginning to be accessible to measurement.

The next decade will tell us things. Perhaps the echoes will appear in the LIGO data — faint, recurring signals after each merger, too consistent to be noise, too strange to fit any standard model. Perhaps the Fermi data will show a small but statistically robust energy dependence in the speed of gamma-ray photons, crossing a threshold that was not crossed in the previous decade's data. Perhaps the tabletop experiments probing quantum gravity will find that the uncertainty principle is not quite what Heisenberg wrote.

Or perhaps they will find nothing. Perhaps the echoes will remain absent, the photon speeds will remain universal, the uncertainty principle will remain unmodified. That, too, is information. That, too, narrows the space of possible theories. Falsification is not failure. It is science doing its job.

What is clear is that we are living through an unusual moment. The instruments are good enough to ask the deepest questions. The frameworks that might answer them — UST among them — are specific enough to be tested. The next generation of gravitational wave detectors, the Einstein Telescope and LISA, will be so sensitive they will see mergers across the entire observable universe. The

next generation of cosmic microwave background telescopes will image the oldest light in the universe with a precision that can probe physics at the Planck scale indirectly.

Somewhere in that coming flood of data is an answer. Whether it is the answer Dustin Lee expects, whether it is something entirely different, or whether it raises still more questions — the way physics always does, at every step — is something we will have to wait to find out.

In the meantime, the laser beams bounce. The mirrors hold their silence. The data accumulates, byte by byte, in server rooms in Louisiana and Washington state, in Pisa and Kamioka, in Chile and the South Pole. The universe is speaking. We are still learning how to listen.

Dustin Lee is a theoretical physics researcher based in Walstonburg, North Carolina. He is the originator of the Unified Substrate Theory (UST), a framework proposing a single underlying field as the common origin of spacetime, matter, and quantum behavior. His work includes the derivation of thirteen experimental predictions distinguishing UST from standard general relativity and quantum field theory.

A note on this article: The Unified Substrate Theory represents an active, developing theoretical framework. Predictions discussed herein are those put forward by the theory's author. Independent peer review and experimental confirmation remain ongoing. This article reflects the state of the theory and the relevant experimental landscape as of May 2026.